

SYTOMINO, Russian Federation

WMO: 238470

Lat: 61.3019N

Lon: 71.3022E

Elev: 110

StdP: 14.64

Time Zone: 5.00 (E05)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-36.3	-46.4	0.4	-40.5	-41.4	0.5	-35.8	18.6	3.4	16.7	6.0	3.4	300	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	83.6	66.4	80.5	65.3	77.2	63.9	69.2	80.0	67.5	77.1	65.5	74.1	5.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	94.0	74.3	63.7	88.5	72.6	62.0	83.3	71.4	33.4	80.0	32.0	77.1	30.5	73.8	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.2	12.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-47.1	71.3	3.4	1.6	-49.6	72.4	-51.5	73.3	-53.5	74.2	-55.9	75.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	-5.7	1.7	15.4	29.3	44.1	61.0	65.8	58.3	46.7	31.5	8.1	0.0
	DBStd	27.27	16.63	16.59	13.17	9.76	10.27	8.45	6.88	6.92	6.94	9.12	14.79	16.88
	HDD50	8574	1728	1353	1074	621	247	12	1	12	142	575	1258	1551
	HDD65	13013	2193	1773	1539	1070	653	174	74	226	548	1039	1708	2016
	CDD50	1213	0	0	0	0	64	342	491	270	44	2	0	0
	CDD65	178	0	0	0	0	5	54	100	19	0	0	0	0
	CDH74	1425	0	0	0	0	52	430	819	124	0	0	0	0
	CDH80	309	0	0	0	0	6	88	202	13	0	0	0	0
Wind	WSAvg	6.0	5.9	5.5	6.3	6.8	6.7	6.2	5.6	5.4	5.8	6.5	6.0	5.7
Precipitation	PrecAvg	30.9	2.0	1.6	2.2	2.1	2.3	3.1	3.0	3.5	2.7	3.1	3.0	2.3
	PrecMax	37.9	3.5	3.1	5.2	3.4	4.9	4.9	8.0	6.0	5.0	5.2	5.3	3.9
	PrecMin	24.3	0.7	0.3	0.2	0.2	0.7	1.8	1.2	1.6	1.1	1.2	1.0	1.0
	PrecStd	3.2	0.8	0.6	1.0	0.7	0.9	0.9	1.3	1.0	0.9	1.0	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	34.6	39.8	54.9	80.4	86.8	89.4	81.6	69.0	56.9	34.9	31.4
		MCWB	31.0	32.3	33.6	45.8	61.7	68.8	70.1	68.1	58.5	48.8	33.1	30.0
	2%	DB	26.6	31.8	36.8	48.8	73.3	82.5	85.5	77.4	63.8	52.1	32.8	28.3
		MCWB	25.4	30.5	32.2	40.8	57.4	65.7	68.2	65.4	55.3	46.7	31.5	27.3
	5%	DB	20.8	27.2	34.5	44.9	67.4	79.1	81.9	74.2	60.1	47.6	30.9	25.2
		MCWB	19.9	25.7	31.2	38.8	55.1	64.2	66.3	63.7	53.7	43.8	29.8	24.2
	10%	DB	16.8	23.0	32.4	41.7	61.2	75.4	79.0	70.7	57.1	43.4	27.5	20.6
		MCWB	15.9	21.7	29.4	36.9	50.8	62.6	64.7	61.7	52.1	40.5	26.2	19.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.2	33.1	34.4	48.3	62.6	70.0	72.3	70.2	59.7	49.8	33.8	30.5
		MCDB	32.3	34.1	37.8	55.9	76.4	82.1	86.0	79.2	67.0	55.5	34.7	31.5
	2%	WB	25.5	30.4	33.1	42.3	58.9	68.1	70.0	67.2	56.8	46.9	31.9	27.3
		MCDB	26.8	31.8	35.9	45.4	71.6	78.8	81.7	74.9	62.4	51.6	32.7	28.1
	5%	WB	20.1	26.0	31.5	39.4	55.1	66.3	68.2	64.6	54.6	44.1	29.8	24.3
		MCDB	20.9	27.3	34.4	44.8	65.9	75.0	78.7	71.6	59.0	47.1	30.9	25.4
	10%	WB	16.0	21.8	29.6	37.2	51.9	64.1	66.5	62.7	52.4	40.9	26.3	19.9
		MCDB	16.8	23.1	32.5	41.5	60.0	73.4	75.8	70.1	56.5	43.0	27.6	20.7
Mean Daily Temperature Range	5% DB	MDBR	14.1	15.2	17.6	15.4	16.6	16.2	16.4	15.7	11.9	8.9	13.4	13.5
		MCDBR	18.1	16.3	15.4	18.8	24.0	21.1	21.3	20.7	17.2	13.5	13.8	12.9
		MCWBR	17.1	15.8	11.9	12.7	12.1	9.2	8.5	10.7	10.2	8.8	13.4	12.4
	5% WB	MCDBR	17.8	16.3	14.5	16.6	22.4	17.6	19.7	18.5	15.7	11.9	14.6	13.1
		MCWBR	17.0	16.0	11.6	11.8	11.7	9.1	8.7	10.7	10.2	8.5	14.2	12.5
Clear-Sky Solar Irradiance	taub	0.255	0.299	0.352	0.376	0.390	0.404	0.420	0.412	0.357	0.328	0.267	0.237	
	taud	1.967	2.013	1.960	2.066	2.278	2.333	2.281	2.268	2.385	2.244	2.029	1.908	
	Ebn at Noon	134	202	231	254	263	261	252	243	238	195	128	83	
	Edh at Noon	14	25	38	42	37	36	37	35	26	20	13	9	
All-Sky Solar Radiation	RadAvg	115	388	896	1399	1581	1684	1625	1139	693	301	117	55	
	RadStd	10	41	58	144	130	138	162	119	79	33	24	4	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page